Money, Banks, and the Bank of Canada

ECON 101 - Macroeconomics

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Chapter Outline

- **10.1** What Is Money, and Why Do We Need It?
- **10.2** How Is Money Measured in Canada Today?
- **10.3** How Do Banks Create Money?
- **10.5** The Quantity Theory of Money

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10.1 Learning Objective

Goal

Define money and discuss the four functions of money.

- Understand what economists mean by money.
- See how money solves problems with barter.
- Learn the **four functions** of money.
- Discuss what can and cannot serve as money.

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What Is Money?

- **Money** is any asset that people are generally willing to accept:
  - in exchange for goods and services, or
  - in payment of debts.
- **Asset:** anything of value owned by a person or a firm.
- Money is one of the most important inventions in economic history because:
  - it makes trade easier,
  - it allows specialization,
  - it supports the development of complex economies.

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Life Without Money: Barter

- Imagine living before money existed.
- All trade would be **barter**:
  - trading goods and services directly for goods and services.
- Barter requires a double coincidence of wants:
  - you must have what I want,
  - and I must have what you want,
  - at the same time and place.
- Barter works in small, simple economies, but breaks down as:
  - the number of goods and services grows,
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- Barter works in small, simple economies, but breaks down as:
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- To avoid the problems of barter, societies adopted **commodity money**.
- **Commodity money:**
  - goods used as money that also have value independent of their use as money.
- Examples:
  - Precious metals such as gold and silver,
  - Beaver pelts among early settlers and some Indigenous peoples in Canada,
  - Cigarettes in prisons and prisoner-of-war camps.
- Commodity money works as long as people:
  - value the commodity itself,
  - and agree to use it in trade.

From Barter to Commodity Money

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The Four Functions of Money

- Money fulfills **four primary functions**:
  1. **Medium of exchange**
  2. **Unit of account**
  3. **Store of value**
  4. **Standard of deferred payment**

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Function 1: Medium of Exchange

- Money is a **medium of exchange** if:
  - it is acceptable to a wide variety of parties
  - as payment for goods and services.
- Instead of barter:
  - you sell your labour or goods $\Rightarrow$ receive money,
  - then use money to buy what you want.
- This greatly reduces the time and effort needed to find trading partners.

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Function 2: Unit of Account

- Money is a **unit of account**:
  - it provides a standard way of measuring the value of goods and services.
- In Canada, prices are quoted in \$ Canadian dollars.
- Without a unit of account:
  - every good would have to be priced in terms of every other good.
  - In a large economy, this would be extremely confusing.

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Function 3: Store of Value

- Money is a **store of value**:
  - it allows people to transfer purchasing power from today to the future.
- Other assets also store value:
  - stocks, bonds, houses, land, art, etc.
- Money is special because it is:
  - highly liquid (easily converted into goods and services),
  - relatively stable in value in low-inflation economies.
- Inflation reduces the effectiveness of money as a store of value.

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Function 4: Standard of Deferred Payment

- Money is a **standard of deferred payment**:
  - it is widely accepted for settling debts that are paid back in the future.
- Loans and credit contracts are typically denominated in money terms.
- This works well when:
  - the value of money is reasonably predictable over time.

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What Can Serve as Money?

- For an asset to serve as a widely accepted medium of exchange, it should:
    1. Be **acceptable** to most people.
    2. Have **standardized quality** so any two units are alike.
    3. Be **durable** so it does not wear out quickly.
    4. Be **valuable relative to its weight**, so it is easy to transport.
    5. Be **divisible** enough to make both small and large transactions.
- Historical and modern monies (gold coins, bank notes, electronic deposits) satisfy these conditions to different degrees.

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Commodity Money: Examples

- **Commodity money** has value independent of its use as money.
- Examples:
  - Gold and silver coins.
  - Beaver pelts in early Canadian trade.
  - Cigarettes used in prisons or war camps.
- Advantage:
  - intrinsic value supports trust in the money.
- Disadvantages:
  - can be costly to store and transport,
  - supply may be hard to control.

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Fiat Money (1 of 2)

- In modern economies, money is typically **paper currency** issued by a central bank.
- In Canada, the central bank is the **Bank of Canada**.
- Modern currencies are **not** convertible into gold or any commodity.
- **Fiat money:**
  - money, such as paper currency, that is authorized by a central bank or government,
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Fiat Money (2 of 2)

- Fiat money is more flexible:
  - central banks can adjust the money supply to respond to shocks.
- But it also has a potential problem:
  - it is valuable only as long as households and firms **have confidence** in it.
- If people lose confidence that paper dollars will hold their value:
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10.2 Learning Objective

Goal

Discuss the definitions of the money supply used in Canada today.

- Different assets vary in how **liquid** they are.
- The Bank of Canada uses several definitions of the money supply:
  - from narrow measures (very liquid) to broad measures (include less liquid assets).

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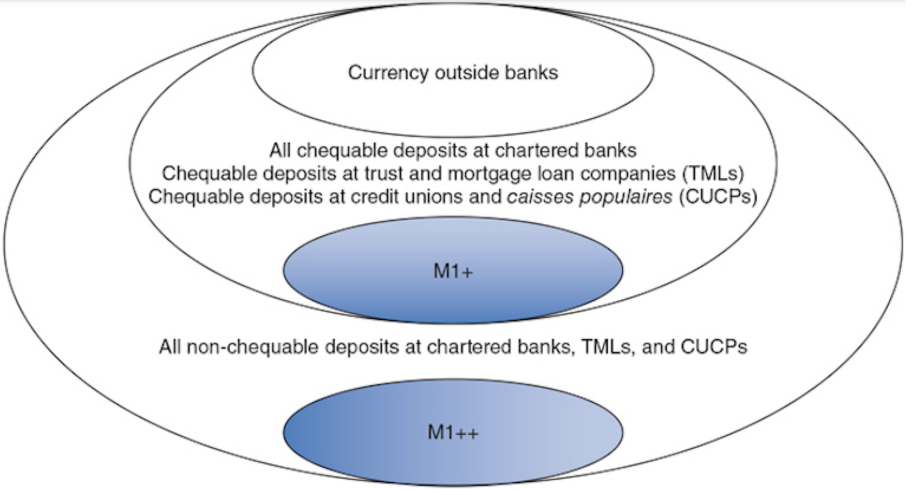
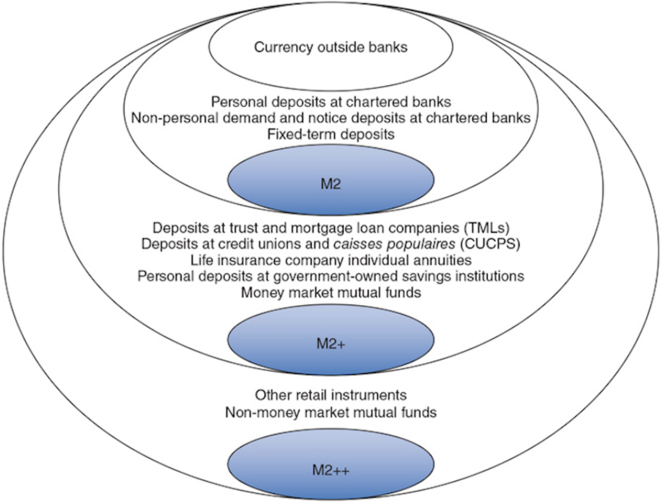
Figure 10.1 – Components of the Money Supply (1 of 3)



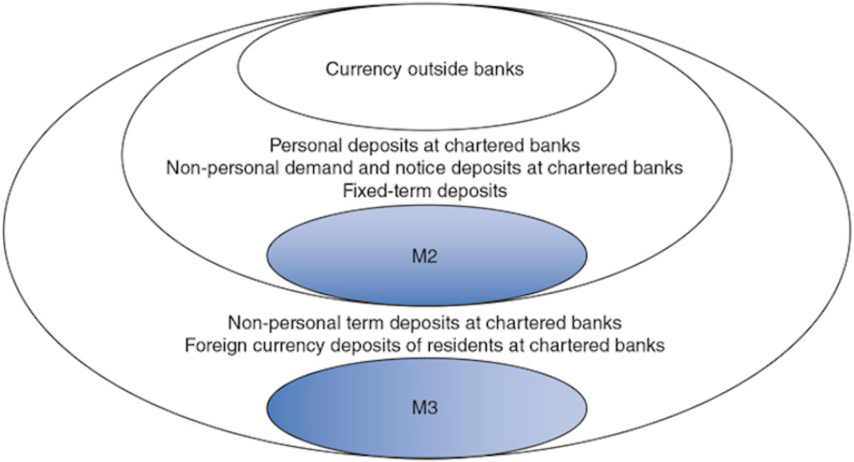
(a) Components of M1+ and M1++

Figure 10.1 – Components of the Money Supply (2 of 3)



(b) Components of M2, M2+, and M2++

Figure 10.1 – Components of the Money Supply (3 of 3)



(c) Components of M3

The Bank of Canada's Measures of Money

- The Bank of Canada uses six main measures:
  - **M1+**, **M1++**, **M2**, **M2+**, **M2++**, **M3**.
- **M1+**: narrowest definition.
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Definition of M1+ and M1++

- **M1+:**
  - Currency in circulation (bank notes and coins outside banks).
  - **Chequable deposits** at:
    - banks,
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- **M2+**: includes M2 plus:
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Which Money Supply Measure to Use?

- Any of the main measures might be relevant, depending on the question.
- For most macroeconomic analysis, we focus on money as a **medium of exchange**.
- That suggests concentrating on **M1+**:
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10.3 Learning Objective

Goal

Explain how banks create money.

- Banks are central to the money creation process.
- More money is held in chequing accounts than exists as physical currency.
- Banks are profit-making private firms; they:
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    - make loans,
    - and thereby create money.

10.3 Learning Objective

Goal

Explain how banks create money.

- Banks are central to the money creation process.
- More money is held in chequing accounts than exists as physical currency.
- Banks are profit-making private firms; they:
  - accept deposits,
  - make loans,
  - and thereby create money.

Figure 10.2 – Balance Sheet for Royal Bank

Asset (in billions)		Liabilities and Shareholders' Equity (in billions)	
Reserves	$199	Deposits	$1082
Loans	950	Borrowing	24
Securities	282	Other liabilities	531
Other assets	302		
		Total liabilities	$1637
		Shareholders' equity	96
Total assets	$1733	Total liabilities and shareholders' equity	$1733

Bank Balance Sheets: Assets and Liabilities

- On a **balance sheet**:
  - Assets are listed on the left,
  - Liabilities and stockholders' equity (**net worth**) on the right.
- **Assets** for a bank include:
  - Reserves (cash in vaults and deposits at the Bank of Canada),
  - Loans,
  - Securities and other investments.
- **Liabilities** include:
  - Deposits (chequing, savings, term deposits),
  - Borrowings.
- The two sides must always sum to the same total.

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Reserves and Desired Reserve Ratio

- **Reserves:**
  - deposits that a bank keeps as cash in its vault or on deposit with the Bank of Canada.
- Banks in Canada are **not** required to hold a specific level of reserves by law.
- They choose to keep a fraction of deposits on hand as reserves:
  - often around 5%.
- **Desired reserve ratio** (r_d):
  - the fraction of deposits banks *want* to keep as reserves.
- Banks may adjust r_d depending on:
  - the state of the economy,
  - and financial market conditions.

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T-Accounts: A Simple Tool

- A **T-account** is a simplified balance sheet.
- It shows only how a transaction changes assets and liabilities.
- Example: you deposit \$1,000 in currency at Bank of Montreal (BMO).
- BMO's T-account change:

Assets	Liabilities
Reserves $+\$1{,}000$	Deposits $+\$1{,}000$

- The currency component of the money supply falls by \$1,000, but chequing deposits rise by \$1,000, so total money supply is unchanged **so far**.

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From Deposits to Loans: Creating New Money

- BMO holds some of the \$1,000 as reserves and lends out the rest.
- Suppose it keeps 10% (\$100) as reserves and lends \$900.
- T-account changes for BMO after the loan:

Assets	Liabilities
Reserves $+\$1{,}000$	Deposits $+\$1{,}000$
Loans $+\$900$	Deposits $+\$900$

- The \$900 loan is typically deposited at another bank, so:
  - the banking system now has:
    - \$1,000 in new reserves,
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Multiple Rounds of Money Creation

- The process continues across banks:
  - each bank keeps 10% of new deposits as reserves,
  - and lends out the remaining 90%.
- First deposit: \$1,000 reserves.
- First loan: \$900 new deposit in the system.
- Second round:
  - 10% of \$900 (i.e., \$90) kept as reserves,
  - \$810 loaned out and deposited.
- Third round:
  - 10% of \$810 kept as reserves,
  - \$729 loaned out, etc.

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When Will It End?

- Each round of lending creates smaller and smaller new deposits.
- With a 10% desired reserve ratio ($r_d = 0.10$):
  - the original \$1,000 in currency becomes the 10% reserves for all the new deposits.
- So total checking deposits created will be:

$$\text{Total deposits} = \frac{1}{r_d} \times \text{new reserves} = \frac{1}{0.10} \times \$1{,}000 = \$10{,}000.$$

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Simple Deposit Multiplier

- The total increase in deposits from an initial increase in reserves can be written as:

$$\Delta\text{Deposits} = \Delta\text{Reserves} + (1 - r_d)\,\Delta\text{Reserves} + (1 - r_d)^2\,\Delta\text{Reserves} + \cdots$$

- This is a geometric series whose sum is:

$$\Delta\text{Deposits} = \frac{1}{r_d} \times \Delta\text{Reserves}.$$

- The **simple deposit multiplier** is:

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Simple Deposit Multiplier: Numerical Examples

- If $r_d = 0.10$ (10% desired reserve ratio):

$$\text{Simple deposit multiplier} = \frac{1}{0.10} = 10.$$

- If $r_d = 0.20$ (20% desired reserve ratio):

$$\text{Simple deposit multiplier} = \frac{1}{0.20} = 5.$$

- Example:
  - New deposits of \$100,000, with $r_d = 0.10$,
  - can support total deposits of:

$$\frac{1}{0.10} \times \$100{,}000 = \$1{,}000{,}000.$$

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- Example:
  - New deposits of \$100,000, with $r_d = 0.10$,
  - can support total deposits of:

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Simple vs. Real-World Deposit Multiplier

- The simple deposit multiplier is an **upper bound**.
- In practice, the real-world deposit multiplier is **smaller** because:
    1. Banks may choose to hold **excess reserves** (more than desired).
    2. Some of the money loaned out is held as **currency** rather than redeposited in banks.
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- In general, the real-world deposit multiplier is greater than 1.
- Therefore:
  - When banks gain reserves, they make new loans, and the money supply expands.
  - When banks lose reserves, they reduce loans, and the money supply contracts.
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- Over these years, prices in Europe rose steadily.
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The Quantity Equation

- In the early 20th century, Irving Fisher formalized this relationship in the **quantity equation**:

$$M \times V = P \times Y,$$

where:

  - M = money supply,
  - V = velocity of money,
  - P = price level,
  - Y = real output (real GDP).
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Calculating the Velocity of Money

- Rearranging the quantity equation:

$$V = \frac{P \times Y}{M}.$$

- If we measure:
  - M with M1+,
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- then we can compute V for any year.
- In practice, V fluctuates from year to year, but it is often fairly stable over longer periods in some economies.

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Quantity Theory of Money

- The **quantity theory of money** assumes that the velocity of money is approximately constant in the long run.
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From Levels to Growth Rates

- When variables are multiplied in an equation, we can write an equivalent relationship in terms of their **growth rates**.
- Let:
  - g_M = growth rate of money,
  - g_V = growth rate of velocity,
  - π = inflation rate (growth rate of the price level P),
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- Then the quantity equation implies:

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Inflation According to the Quantity Theory

- If we assume that V is constant in the long run:

$$g_V = 0,$$

- then the growth-rate form simplifies to:

$$g_M = \pi + g_Y.$$

- Solving for inflation:

$$\pi = g_M - g_Y.$$

- Predictions:
  - If $g_M > g_Y$: $\pi > 0$ (inflation).
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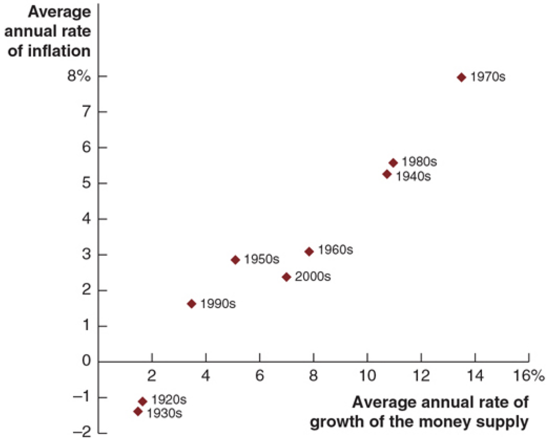
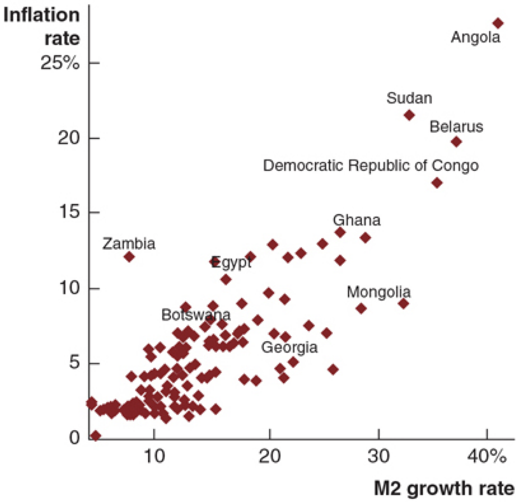
Figure 10.7 – Money Growth and Inflation (1 of 2)



(a) Inflation and money supply growth in Canada, 1920s–2010s

Figure 10.7 – Money Growth and Inflation (2 of 2)



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- Example:
  - Zimbabwe experienced hyperinflation so severe that people stopped using the national currency.
- Hyperinflation is usually associated with:
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Quantity Theory: Main Takeaway

- The quantity theory of money is a **long-run** theory.
- It emphasizes that:
  - persistent inflation is ultimately a monetary phenomenon.
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Summary

- Money is any asset widely accepted in exchange for goods, services, or debt payments.
- Money performs four key functions:
  - medium of exchange,
  - unit of account,
  - store of value,
  - standard of deferred payment.
- Canada's money supply has several measures; we emphasize M1+ for its role as a medium of exchange.
- Banks create money by accepting deposits and making loans, subject to their desired reserve ratio.
- The simple deposit multiplier equals $1/r_d$, but the real-world multiplier is smaller.
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